

Multivariate Random Variables

- Bivariate Random Vector: Given a random experiment with a sample space $\mathcal{C}$. Consider two random variables X_1 and X_2 , which assign to each element c of $\mathcal{C}$ one and only one ordered pair of numbers $X_1(c) = x_1, X_2(c) = x_2$. Then, we say that (X_1, X_2) is a random vector. The space of (X_1, X_2) is the set of ordered pairs $\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathcal{C}\}$.
- The above definition can be extended into a random vector with any dimension.

- Multinomial distribution: $X = (X_1, \dots, X_k) \sim \text{multi}(n; p_1, \dots, p_k)$
- Multivariate normal distribution: $X \sim N(\mu, \Sigma)$

Expectation of the Random Vector

- The concept of expectation extends in a straightforward manner.
- Let (X_1, X_2) be a random vector and let $Y = g(X_1, X_2)$ for $g : R^2 \rightarrow R$. Find $\mathbb{E}[Y]$.
- Suppose (X_1, X_2) is of the continuous type.

$$\mathbb{E}(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_1, x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

- Suppose (X_1, X_2) is of the discrete type.

$$\mathbb{E}(Y) = \sum_{x_1} \sum_{x_2} g(x_1, x_2) p_{X_1, X_2}(x_1, x_2)$$

- **Theorem.** Let (X_1, X_2) be a random vector. Let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables whose expectation exist. Then for any real number k_1 and k_2 ,

$$\mathbb{E}(k_1 Y_1 + k_2 Y_2) = k_1 \mathbb{E}(Y_1) + k_2 \mathbb{E}(Y_2).$$

- Note that the expected value of any function $g(X_2)$ of X_2 can be found in two ways:

$$\mathbb{E}(g(X_2)) = \int_{-\infty}^{\infty} g(x_2) f_{X_2}(x_2) dx_2$$

$$\mathbb{E}(g(X_2)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

- mgf: Let $X = (X_1, X_2)'$ be a random vector.

$$M_{X_1, X_2}(t_1, t_2) = \mathbb{E}(e^{t_1 X_1 + t_2 X_2})$$

Let $t = (t_1, t_2)'$, then

$$M_{X_1, X_2}(t_1, t_2) = \mathbb{E}(e^{t'X})$$

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$$M_{X_1}(t_1) = M_{X_1, X_2}(t_1, 0)$$

$$M_{X_2}(t_2) = M_{X_1, X_2}(0, t_2)$$